Group II Row IV: from the three lines to JIMWLK, BFKL, DGLAP, running coupling and finite terms

1 Notation

- $d_\perp = 2-\epsilon$; $a \equiv \frac{k^+}{\vee}$; $X \equiv y'-y$; $c_0 = 2e^{-\gamma_E}$; $\hat{\mu}^2 \equiv \frac{\mu_{\overline{MS}}^2 e^{2\gamma_E}}{4} = \frac{\mu_{\overline{MS}}^2}{c_0^2}$; $\beta_0 = \frac{11}{3}N_c$.
- $c \equiv \frac{g^2 N_c}{4\pi^2} = \frac{g^2}{8\pi^2}\,2N_c = \frac{\alpha_s}{2\pi}2N_c$; $\mathcal{K}(r) \equiv \frac{r}{r^2}$.
- LO with a weight $f(X)$ *inside* the integral, and its rescaled version:

$$\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\, f(y'-y)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} \times \big[U^{e'c}(y') - U^{e'c}(x')\big]\big[U^{ec}(y) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \qquad \frac{d^3N_{\mathrm{LO}}}{d^3k}[f] \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[f]\Big|_{\xi=1}. \tag{1.1}$$

- $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}}$: Eq. (1.1) with $f=1$ and the kernel of line 3.
- The point-z LO (both gluon points at z, no phase) and H_4:

$$\frac{d^3N_{\mathrm{LO}}}{d^3k}^{y} \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,x',z} \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x), \tag{1.2}$$

$$H_4 \equiv \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \times \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x). \tag{1.3}$$

2 The starting point

Your Row IV, the three lines (L1), (L2), (L3):

$$\begin{aligned}
(\mathrm{L1}) &= c\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big]\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}_0],\\
(\mathrm{L2}) &= c\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big)\Big]\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}_0],\\
(\mathrm{L3}) &= c\int_a^{\frac{k^+}{k^++\Lambda}} d\xi\,\frac{1-\xi}{\xi}\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}}\Big|_{d_\perp},
\end{aligned} \tag{2.1}$$

with

$$\mathcal{B}_0 \equiv -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{X^2\mu_{\overline{MS}}^2}{4} - 1. \tag{2.2}$$

The bracket depends on X, so it is a weight inside the integrals.

3 Step 0: what line 2 is

Put $p^+ = k^+\frac{1-\xi}{\xi}$. Then $\xi = 1 \leftrightarrow p^+ = 0$, $\xi = a \leftrightarrow p^+ = \vee - k^+$, $\xi = \frac{k^+}{k^++\Lambda} \leftrightarrow p^+ = \Lambda$, and

$$\frac{d\xi}{\xi^2}\,\frac{\xi}{1-\xi} = \frac{dp^+}{p^+}. \tag{3.1}$$

Write $F(\xi) \equiv \frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}[\mathcal{B}_0]$, so $F(1) = \frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}_0]$. The $\frac{\xi}{(1-\xi)_+}$ term of (L1) is your p^+ prescription:

$$\int_a^1 \frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}F \equiv \int_a^1 d\xi\,\frac{F(\xi)-F(1)}{\xi(1-\xi)} = \int_0^{\vee-k^+} dp^+\,\frac{F(p^+)-F(0)}{p^+}. \tag{3.2}$$

The unregulated integral with the cut $p^+ \geq \Lambda$ is, exactly,

$$\int_\Lambda^{\vee - k^+} \frac{dp^+}{p^+} F = \int_0^{\vee - k^+} dp^+ \frac{F(p^+) - F(0)}{p^+} + F(0)\Big[\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+} - 1\Big)\Big] + O\Big(\frac{\Lambda}{k^+}\Big). \tag{3.3}$$

So (L2) is the subtraction term of your plus prescription: (L1)+(L2) is the original integral over $p^+ \geq \Lambda$. (L2) has the LO Wilson-line structure because $F(0) = \frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}_0]$. (Checked: 4×10^{-16}, `g2iv_full/plus_check.py`.)

4 Step 1: the ϵ-bracket

With $d_\perp = 2 - \epsilon$, $-\frac{d_\perp}{\epsilon} = -\frac{2}{\epsilon} + 1$, and the -1 cancels. Exactly:

$$\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\Big[2\gamma_E + \log\frac{X^2\mu^2_{\overline{MS}}}{4}\Big] = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ln\big(X^2\hat{\mu}^2\big) \equiv \mathcal{B}. \tag{4.1}$$

Where no pole multiplies it, $\frac{d_\perp}{2} = 1 - \frac{\epsilon}{2} \to 1$.

5 Step 2: standard plus distribution; the $\log(\frac{\vee}{k^+} - 1)$ cancels

DGLAP uses the standard plus distribution on $[0,1]$, restricted to $[a,1]$:

$$\int_a^1 \frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}\Big|_{\rm std} F \equiv \int_a^1 d\xi\,\frac{F(\xi)/\xi - F(1)}{1-\xi} + F(1)\log(1-a). \tag{5.1}$$

The difference from yours, Eq. (3.2):

$$\begin{aligned}\text{yours} - \text{std} &= \int_a^1 d\xi\Big[\frac{F - F(1)}{\xi(1-\xi)} - \frac{F/\xi - F(1)}{1-\xi}\Big] - F(1)\log(1-a) = F(1)\int_a^1 d\xi\Big[-\frac{1}{\xi(1-\xi)} + \frac{1}{1-\xi}\Big] - F(1)\log(1-a)\\ &= -F(1)\int_a^1 \frac{d\xi}{\xi} - F(1)\log(1-a) = -F(1)\Big[\log\frac{\vee}{k^+} + \log\Big(1 - \frac{k^+}{\vee}\Big)\Big] = -F(1)\log\Big(\frac{\vee}{k^+} - 1\Big)\end{aligned} \tag{5.2}$$

(checked: 6×10^{-17}). Define the standard splitting function

$$\mathcal{P}_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+}\Big|_{\rm std} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big] + \frac{\beta_0}{2}\delta(1-\xi), \tag{5.3}$$

with $\delta(1-\xi)$ at the endpoint counted fully: $\int_a^1 \frac{d\xi}{\xi^2}\delta(1-\xi)F = F(1)$. Since $c = \frac{g^2}{8\pi^2}2N_c$:

$$\text{(L1)} = \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi) - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\,\rm LO}}{d^3k}[\mathcal{B}] \;-\; c\log\Big(\frac{\vee}{k^+} - 1\Big)\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]. \tag{5.4}$$

Adding (L2), the $\log(\frac{\vee}{k^+} - 1)$ cancels exactly (both its $\log\frac{\vee}{k^+}$ and its $\log(1 - \frac{k^+}{\vee})$):

$$\boxed{\text{(L1)} + \text{(L2)} = \underbrace{c\,\log\frac{k^+}{\Lambda}\,\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]}_{\text{JIMWLK}} + \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi) - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\,\rm LO}}{d^3k}[\mathcal{B}]} \tag{5.5}$$

- The $\log\frac{\vee}{k^+}$ with the LO Wilson-line structure is *not* a BFKL term: it is gone. (In your split it was the pair $\pm c\log\frac{\vee}{k^+}\frac{d^3N_{\rm LO}}{d^3k}[\mathcal{B}]$.)
- The $\log\frac{k^+}{\Lambda}$ term is the JIMWLK term of this row, as you say.

6 Step 3: DGLAP and running coupling

Split Eq. (5.5) at $\frac{\beta_0}{2}\delta(1-\xi)$:

$$\underbrace{-\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2)\Big]}_{\rm (D)} + \underbrace{\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\rm LO}}{d^3k}\Big[\frac{2}{\epsilon} + \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2)\Big]}_{\rm (R)}. \tag{6.1}$$

DGLAP. The hadron is $\frac{d^3N_h}{d^3k} = \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3} D_{h/g}(\zeta) \frac{d^3N_g}{d^3k}\big(\frac{k^+}{\zeta}, \frac{\mathbf{k}}{\zeta}\big)$, and $D^{\rm bare} = D(\mu_F) + \frac{g^2}{8\pi^2}\big[\frac{2}{\epsilon} + \ln\frac{\mu^2}{\mu_F^2}\big]\mathcal{P}_{gg}\otimes D$.

- The counterterm covers the same region as the ξ integral ($a \le \zeta\xi \le \zeta \le 1$), with the same standard plus distribution. So the pole of (D) cancels.
- What is left is

$$\text{(D)} \to -\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\ln\frac{X^2\mu_F^2}{c_0^2}\Big]. \tag{6.2}$$

Running coupling. $\frac{d^3N_{\rm LO}}{d^3k} \propto g^2$, and the $\overline{MS}$ coupling counterterm adds $-\frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\frac{d^3N_{\rm LO}}{d^3k}$:

$$\text{(R)} - \frac{g^2\beta_0}{16\pi^2}\frac{2}{\epsilon}\frac{d^3N_{\rm LO}}{d^3k} = \frac{\alpha_s\beta_0}{4\pi}\frac{d^3N_{\rm LO}}{d^3k}\big[\ln(X^2\hat\mu^2)\big], \qquad \frac{d^3N_{\rm LO}}{d^3k}\Big|_{\alpha_s} + \frac{\alpha_s\beta_0}{4\pi}\frac{d^3N_{\rm LO}}{d^3k}\big[\ln(X^2\hat\mu^2)\big] = \frac{d^3N_{\rm LO}}{d^3k}\Big|_{\alpha_s\to\alpha_s(X^2)}, \tag{6.3}$$

with $\alpha_s(X^2) = \alpha_s + \frac{\alpha_s^2\beta_0}{4\pi}\ln(X^2\hat\mu^2) + O(\alpha_s^3)$ and $X = y' - y$ (checked, sympy).

7 Step 4: the BFKL term hidden in (D)

At small ξ, $\mathcal{P}_{gg} \to 2N_c\frac{1-\xi}{\xi}$, and the y' integral is a Fourier transform at $\mathbf{k}/\xi$. Write the integrand of $\frac{d^3N_{\xi\,\rm LO}}{d^3k}[f]$ as $e^{-ik\cdot(y'-y)/\xi} f(y'-y)\,F_y(y')$, with F_y smooth.

1. The constants and $\ln\mu^2$ in the weight multiply $\int d^2y' e^{-ik\cdot(y'-y)/\xi} = (2\pi\xi)^2\delta^2(\mathbf{k}) = 0$ for $\mathbf{k}\neq 0$.
2. $\nabla^2\ln X^2 = 4\pi\delta^2(X)$ gives $\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\ln(y'-y)^2 = -\frac{4\pi\xi^2}{\mathbf{k}^2}$ (checked).
3. Hence $\int d^2y'\, e^{-ik\cdot(y'-y)/\xi}\ln(X^2\hat\mu^2)F_y(y') = -\frac{4\pi\xi^2}{\mathbf{k}^2}F_y(y)\big[1+O(\xi)\big]$. Numerically (`advisor/smallxi.py`) the ratio to the leading term is 1.008, 1.002, 0.99994, 0.99979 at $\xi = 0.2, 0.1, 0.05, 0.03$.
4. At $y' = y \equiv z$: $\frac{(x'-y')}{(x'-y')^2} \to \mathcal{K}(x'-z)$ and $U(y') \to U(z)$. So $F_y(y)$ is the integrand of $\frac{d^3N_{\rm LO}}{d^3k}^{y}$, with the Wilson-line structure $[U(z)-U(x')][U(z)-U(x)]$.

The small-ξ end of (D) is therefore

$$-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\frac{d^3N_{\rm LO}}{d^3k}^{y} = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{y}\int_a^1 d\xi\,\frac{1-\xi}{\xi} = \Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4, \tag{7.1}$$

since $\int_a^1\frac{1-\xi}{\xi}d\xi = \log\frac{1}{a} - (1-a)$ and $H_4 = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{y}$. The last identity holds because $\int_w e^{ik\cdot w}\mathcal{K}^i(w-z) = \frac{2\pi i k^i}{\mathbf{k}^2}e^{ik\cdot z}$, so $\int_{w,w'} e^{-ik\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z) = \frac{4\pi^2}{\mathbf{k}^2}$ (finite, no $\log\mu$; checked).

- Expanding $[U(z)-U(x')][U(z)-U(x)]\rho\rho$ gives exactly your four-line BFKL term ($\frac{g^4}{16\pi^5}\cdot 2N_c = \frac{g^4N_c}{8\pi^5}$).
- **This is where your BFKL term comes from**, with the right Wilson-line structure. It is the parent gluon with large p^+ (BFKL region) inside the DGLAP integral, and no other row has this structure.

8 Step 5: extract it

Add and subtract Eq. (7.1) inside (D):

$$\text{(D)} = \underbrace{-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat\mu^2)\Big] + \frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\frac{d^3N_{\rm LO}}{d^3k}^{y}\Big\}}_{\text{(D}_{\rm sub}\text{): DGLAP with the BFKL region subtracted}}$$
$$\underbrace{+\log\frac{\vee}{k^+}H_4}_{\to\ \text{BFKL}}\ \underbrace{-\Big(1-\frac{k^+}{\vee}\Big)H_4}_{\text{finite}}. \tag{8.1}$$

- By Step 4 the integrand of (D$_{\rm sub}$) is integrable at $\xi\to 0$, so (D$_{\rm sub}$) has no $\log\frac{\vee}{k^+}$.
- The subtraction has no pole and no μ: the DGLAP renormalization of Step 3 is unchanged.
- Everything is symmetric between amplitude and conjugate. $\frac{d^3N_{\rm LO}}{d^3k}^{y}$ has both gluon points at one z.
- This replaces the $-2N_c\frac{1-\xi}{\xi}\frac{d^3N^y_{\xi\rm LO}}{d^3k}[\log\ldots]$ of the earlier Eq. (5.12), which as written (log outside) is zero.

9 Step 6: line 3

$\int_a^{\frac{k^+}{k^+ + \Lambda}} = \int_a^1 + O(\frac{\Lambda}{k^+})$, since the integrand $\propto (1-\xi)$. At small ξ, $\frac{d^3 N_{\xi\,\mathrm{LO}}}{d^3k}^{\mathrm{lp}} = O(\xi^2)$ (no $\delta^2(\mathbf{k})$ term), so there is no $\log \frac{\vee}{k^+}$. No pole multiplies it, so $d_\perp \to 2$:

$$(\mathrm{L3}) = c \int_a^1 d\xi \, \frac{1-\xi}{\xi} \int_{x,y,x',y'} \ldots \Big[\mathcal{K}(x-y) \cdot \mathcal{K}(x'-y') - 2 \frac{(x-y)\cdot X \, (x'-y')\cdot X}{(x-y)^2 (x'-y')^2 X^2} \Big] \ldots \quad \text{(finite)}. \tag{9.1}$$

10 Result

$$\begin{aligned}
\text{Row IV} = {} & \underbrace{c \log \frac{k^+}{\Lambda} \frac{d^3 N_{\mathrm{LO}}}{d^3 k} \Big[-\frac{2}{\epsilon} - \ln(X^2 \hat{\mu}^2) \Big]}_{\text{JIMWLK}} + \underbrace{\log \frac{\vee}{k^+} H_4}_{\text{BFKL}} \\
& - \frac{g^2}{8\pi^2} \int_{k^+/\vee}^1 \frac{d\xi}{\xi^2} \Big\{ \mathcal{P}_{gg} \frac{d^3 N_{\xi\,\mathrm{LO}}}{d^3 k} \Big[\frac{2}{\epsilon} + \ln(X^2 \hat{\mu}^2) \Big] + \frac{8\pi N_c \, \xi(1-\xi)}{\mathbf{k}^2} \frac{d^3 N_{\mathrm{LO}}}{d^3 k}^{y} \Big\} \quad \text{(DGLAP)} \\
& + \frac{g^2 \beta_0}{16\pi^2} \frac{d^3 N_{\mathrm{LO}}}{d^3 k} \Big[\frac{2}{\epsilon} + \ln(X^2 \hat{\mu}^2) \Big] \quad \text{(running coupling)} \\
& - \Big(1 - \frac{k^+}{\vee} \Big) H_4 + c \int_{k^+/\vee}^1 d\xi \, \frac{1-\xi}{\xi} \frac{d^3 N_{\xi\,\mathrm{LO}}}{d^3 k}^{\mathrm{lp}} \Big|_{d_\perp = 2} \quad \text{(finite)}
\end{aligned} \tag{10.1}$$

After renormalization: DGLAP $\to \ln \frac{X^2 \mu_F^2}{c_0^2}$ (pole in $D_{h/g}$), and running coupling $\to \frac{d^3 N_{\mathrm{LO}}}{d^3 k}|_{\alpha_s \to \alpha_s(X^2)}$.

Check. The three H_4 structures (BFKL, subtraction, finite) add up to zero:

$$\log \tfrac{\vee}{k^+} H_4 - \big[\log \tfrac{\vee}{k^+} - (1-a) \big] H_4 - (1-a) H_4 = 0,$$

so the result returns to the starting lines (L1)+(L2)+(L3).

piece of your three lines	goes to	Wilson-line structure
$\log \frac{k^+}{\Lambda}$ of (L2)	JIMWLK	LO
$\log(\frac{\vee}{k^+} - 1)$ of (L2)	cancels with your plus prescription (Step 2)	LO
(L1), small-ξ end	BFKL: $+\log \frac{\vee}{k^+} H_4$	$[U(z) - U(x')][U(z) - U(x)]$ (your term)
(L1), rest	DGLAP + running coupling; $-(1 - \frac{k^+}{\vee}) H_4$ finite	LO_ξ
(L3)	finite (linear polarization)	LO_ξ

At leading power ($a \to 0$): $\int_a^1 \to \int_0^1$, $(1-a) H_4 \to H_4$. Checks: `g2iv_full/plus_check.py`, `advisor/checks.py`, `advisor/smallxi.py`.